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REVIEW ARTICLE

Toxicity of cigarette butts and possible recycling solutions—a literature review

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Abstract

Tobacco is a worldwide-consumed product, which in addition to causing public health-related issues is responsible for the most common form of litter in the world—smoked cigarette butts (CBs). A large attention has been drawn to this question, since this specific waste type tends to end up in terrestrial and aquatic ecosystems, posing serious threat to a range of life forms. Decomposition may take several years to occur because cellulose acetate is hardly accessible, before deacetylation, by bacteria and fungi. This review concerns the toxicity derived from smoked cigarette butts, as well as innovative ecological solutions for solving the CB litter problem. Toxicity studies have demonstrated the critical influence of chemicals present in smoked CBs to the environment as a whole, but also the physical contaminating potential considering micro- and nanoparticles derived from CB material. Nevertheless, several technological approaches were aimed at unveiling hidden value within used CBs, as well as propositions for incorporation of this residue in large volume production items or direct recycling. In summary, several methods are available to alleviate CB pollution, while appropriate and efficient collection logistics by consumers appears as the main bottleneck for an effective recycling. It is also clear that while considerable progress has been made recently in light of CB recycling solutions, there is still a vast research capacity in this regard.

Keywords Cigarette butt · Toxicity · Decomposition · Recycling waste · Pollution · Debris

Introduction

Direct disposing of non-degradable residues into the environment is a major pollution issue worldwide. During their existence, these residues have the potential to harm nature in a myriad of ways. Among them, cigarette butts (CBs) are an important part of urban litter because of their massive numbers and of their longevity. According to the 2019 World Health Organization Report on the Global Tobacco Epidemic (WHO Report on the Global Tobacco Epidemic 2019), an estimated 1.1 billion people of 15 years of age and over are considered tobacco users, of which 1.07 billion are considered smokers. Interestingly, this number has declined since the first estimate in 2007, when 1.46 billion people were considered tobacco users. It is easy to imagine the massive amounts of cigarette butts produced over time. Indeed, according to the 2018 Tobacco Atlas (Drope et al. 2018), in 2016 only, about 5.7 trillion cigarettes were smoked worldwide. Public health issues associated with tobacco use and smoking habits have been extensively researched and discussed, and will be not addressed in this paper, but it is worth noting that over 7.1 million deaths were associated with tobacco use in 2016 alone (Drope et al. 2018).

A large proportion of these cigarettes, once smoked, is simply thrown into the ground in the form of a CB, which can still retain unburnt or partially burnt tobacco; eventually, this litter may end up in marine environments as well (Loizidou et al. 2018). Even if a CB has no remaining tobacco, toxic chemicals derived from the smoking process remain trapped in the butt. Moreover, CBs pose serious environmental threats because of their resistance to biodegradation in

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natural environments (Bonanomi et al. 2015; Bonanomi et al. 2020; Joly and Coulis 2018). Standard CBs are composed of plasticized cellulose acetate, compressed and externally wrapped with a paper layer (Fig. 1a, b). In fact, cellulose acetate is a material that needs a deacetylation step to initiate biodegradation process (Puls et al. 2011), and therefore requires the addition of other components to enhance natural degradation.

Bonanomi et al. (2015, 2020) performed extensive research on this matter, using artificially smoked CBs and different environment conditions (laboratory and open field) to evaluate mass loss and CB carbon quality alteration for certain time intervals. They reported that CB decomposition involves three stages: an initial fast decay of 15% to 20% mass loss in the first 30 days, attributed to the degradation of external involving paper; then a slow 2-year decomposition period which accounts for approximately 10% mass loss and is rather independent of surrounding conditions; and finally the last stage where environmental conditions affect decomposition rate significantly (Bonanomi et al. 2020). Considering all conditions tested, CB mass loss after 5 years ranged from 50 to 80%. These results prove that CBs are both short- and long-term toxicity threats because leaching occurs rapidly when CBs are wetted but also occurs over a long period with periodic wettings and slow decomposition.

As an alternative for common cellulose acetate filters, companies have produced filters made only of cellulose, which in theory should degrade much faster. However, according to a study published by Joly and Coulis (2018), the environment in which decomposition takes place influences its rate and Green et al. (2020) reported toxic effects in invertebrates caused by biodegradable CBs. Thus, the substitution for these biodegradable filters should be accompanied by user knowledge and responsibility concerning disposal habits. Robertson et al. (2012) approached this issue from a different perspective. Since the hydrolysis of cellulose acetate into cellulose and acetic acid is a process which limits the decomposition rate, these authors proposed the use of encapsulated, controlled-release acid catalysts to accelerate the process, once they are exposed to environmental water.

The contamination dynamics of littered CBs was studied in 2014 by Roder Green et al. (2014). The authors observed the characteristics of butt littering in the city of Berlin. Because of factors such as preferential littering (in restaurants, or

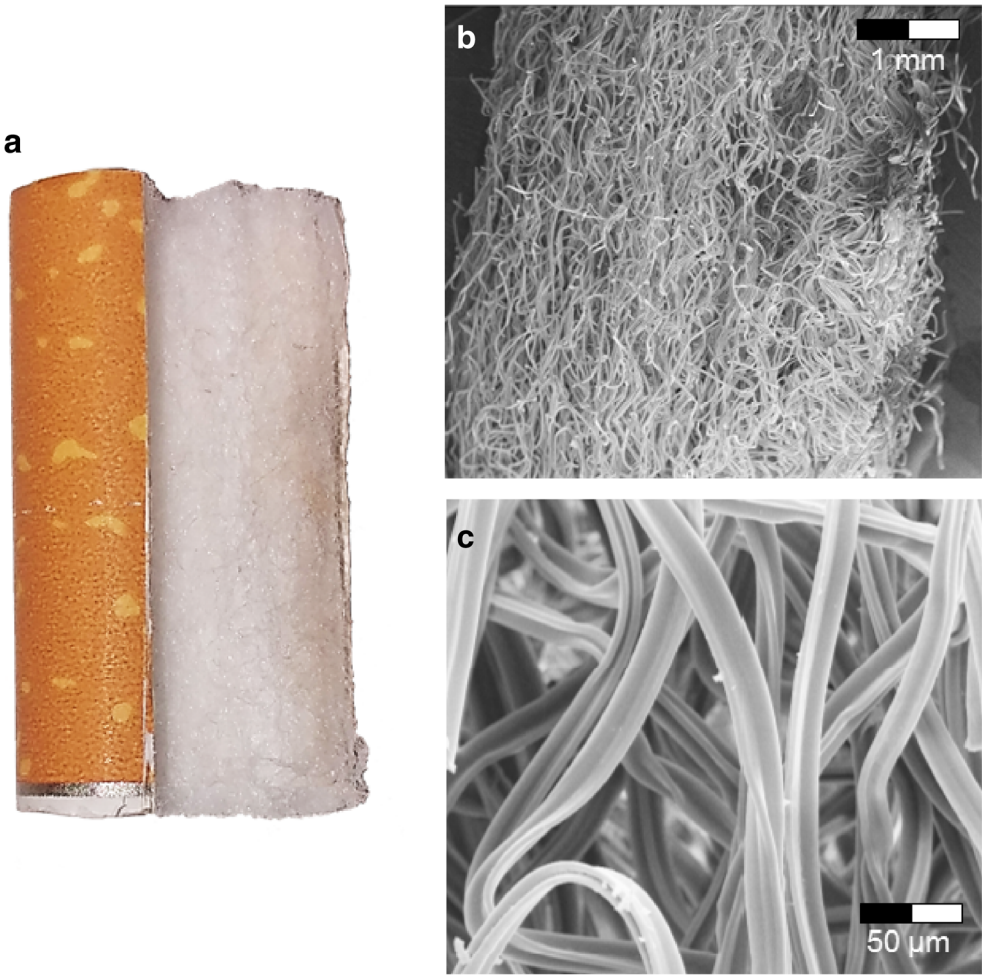


Fig. 1 Structure of CB at different scales. **a** Image of a CB with the external layer made of cellulose and the white, internal filter of cellulose acetate. **b**, **c** Scanning electron microscope (SEM) images of smoked but undecomposed CB at two levels of magnification. Pictures by Giuliano Bonanomi and Antonio Crasto

shopping mall entrances), redistribution processes (cleaning by shop owners, for example), and accumulation in spots not attended by mechanical cleaning, it was found that littering is heterogeneous: average CB concentration was 2.7 butts m^{-2}, while the maximum concentration found was 48.8 butts m^{-2}. The elution kinetics of nicotine in water was also a target of attention. The authors proposed two situations: one, simulating a puddle of standing water, containing cigarette butts, and the other, simulating a rainfall experience over CBs, in which there is the influence of water movement. In the first situation, one CB was added to 1 L of purified water. After several different time intervals, aliquots were taken and nicotine content was measured. Twenty-four-hour exposure resulted in a total release of 7.1 mg g^{-1} of nicotine, a value 14 times higher than the 0.5 mg g^{-1} threshold established by the European Union for hazardous tobacco waste. Also, the dynamics of elution showed that 50% of all the nicotine was released in 27.2 min, and 90% after 10 h. This fact is alarming because it stresses how fast this contamination occurs. The rainfall experiment consisted of placing five CBs on a 0.16-m^2 stainless steel tray, positioned with a 2% slope (equivalent to common sidewalk slopes). A total of 15 daily simulated rainfall events were conducted to meet an estimate of the minimum quantity of water that comes in contact with littered CBs due to rainfall. Results showed a total release of 3.8 mg g^{-1} of nicotine, after 15 events that totalled a precipitation of 21.6 mm; 50% of nicotine release occurred after 1.6 mm of precipitation, equivalent to only 1.1 rainfall events. Finally, after the 15 rainfall events, the remaining CBs were subjected to the conditions of the standing water experiment. Surprisingly, even after leaching nicotine during the rainfall experiments, these CBs still leached 0.16 mg g^{-1} of nicotine after 24-h exposure to standing water, still above threshold limits for the EU, showing that even CBs which were "washed" by rainfall are still an ecological threat to aquatic environments.

The creation of value for CBs plays a role in reducing reckless disposal of this item, and therefore contributes to prevent environmental degradation. According to Torkashvand and Farzadkia (2019), three strategies are mainly used for controlling CB littering: establishment of fines and waste tax; creating responsibility for consumers and producers; and raising awareness about the disadvantages of CBs. If these residues were to be considered a source material with possible financial return, an additional driving force for CB collection would be created almost as a "fourth strategy" of CB littering control.

One can conclude that the permanence of CBs packed with toxic chemicals in the environment is a serious issue, with urgency to its resolution. In addition to the secondary chemicals present in cigarettes and cellulose acetate litters, also nicotine and many other chemical compounds can cause toxic effects on diverse organisms as well. Here, we review available studies that investigate the toxic effect of CBs on a range of organisms and the possible solutions for effective recycling of this waste.

Materials and methods

Literature search and data collection

The search protocol involved articles published up to October 2020, concerning the toxic effect of CBs and potential recycling solutions. The search was carried out in all international journals using online versions of the Web of Science, Science Direct, and Scopus. Keywords used were "cigarette butt", "cigarette litter", and "cigarette filter", and the search was confined to the article title taking care to include also plural variations (butts and filters). Criteria for article selection were decided a priori to avoid personal bias; only articles containing quantitative data on CBs toxicity were included. Basically, the screening process was comprised of the following stages:

(1) Title and abstract: a simple evaluation of both title and abstract was carried out for every result, in relation to cigarette butt toxicity and/or possible applications for cigarette butts to exclude non-related results;
(2) Abstract: articles filtered in (1) had their abstract fully read, in order to further avail the real interest for this review article and filter out non-applicable studies;
(3) Full content: results filtered from (2) had their full content read and evaluated;
(4) Extensive evaluation of every document present in "References" of articles filtered from (2).

Results and discussion

A large number of results was returned for the defined title search strings: "cigarette butt" resulted in approximately 300 articles, "cigarette litter" was related to 12 papers and "cigarette filter" totalled over 400 results. After completion of the screening process, research was focused down to 87 references, 29 of which related to cigarette butt toxicity and 58 associated with potential recycling applications of this waste.

Cigarette butt toxicity

There are different ways in which CBs threaten our environment and living organisms. A summary of the research articles described in the following paragraphs, concerning CB toxicity, is given in Table 1 for purposes of easiness of observation and comparison.

Table 1 List of studies reporting ecotoxicological data of cigarette butts

CB type	Extraction methods	Concentration range	Tested organisms	Main results	Ref.
Tobacco and cigarette butts	No extraction—direct placement in soil and water spraying	Tobacco—0 to 25 g/m^2; CBs—1 to 5/m^2	Parsley, coriander, basil, peppermint	- Every plant subjected to tobacco reached the nicotine maximum residue level (MRL) of 0.01 mg[a] nicotine per kg in the first week; - Concentration as low as 1 CB/m^2 was enough to contaminate basil, peppermint, and coriander over the MRL.	Selmar et al. 2018
Smoked and unsmoked filters, without remnant tobacco	Mixing with seawater and incubation for 96 h	One filter per 40 ml solution.	Diverse bacterial phyla	- Smoked filter addition caused a depletion in *Cyanobacteria* and heterotrophic *Bacteroidetes*, while heterotrophic *Gammaproteobacteria*, *Firmicutes*, and *Thermotogae* showed an enrichment after the experiment.	Quéméneur et al. 2020
Regular and menthol cigarette butts, unsmoked, smoked without remnant tobacco, and smoked with ~ 1-cm remnant tobacco	No extraction—direct placement in mesocosm	1 CB per mesocosm, equivalent to 61 CBs/m^2	Perennial ryegrass (graminae: *Lolium perenne* L.) and white clover (leguminosae: *Trifolium repens* L.)	- Shoot length and germination success were significantly impaired by exposure to any type of CB, for both species; - CBs caused increase in chlorophyll-a, for clover shoots, and in chlorophyll-b for ryegrass shoots; - Chlorophyll-a to -b ratio was increased for clover, and decreased for ryegrass.	Green et al. 2019
Regular cigarette butts, smoked with ~ 1-cm remnant tobacco of nineteen filtered cigarette types	Soaking in solution, with orbital shaker, for 24 h	Geometrically increasing toxicant concentrations, up to 4 CBs/l	Cladoceran (*Ceriodaphnia* cf. *dubia*) and Microtox (*Vibrio fischeri*)	- CBs showed EC50 (immobilisation—48 h) varying from 8.9 to 25.9 mg butts/l to Cladoceran; - CBs showed EC50 (bioluminescence—48 h) varying from 104 to 832 mg butts/l to Microtox.	Micevska et al. 2006
Regular cigarette butts, artificially smoked with ~ 1-cm remnant tobacco	Soaking in drinking water overnight, at room temperature	0.023 CB/l and 0.23 CB/l	Freshwater mussel (*Anodontites trapesiali*)	- Although there was no immune response suppression or morphometric changes in the haemocytes, nuclear abnormalities were induced.	Montalvão et al. 2019b
Regular cigarette butts, artificially smoked with remnant tobacco	Soaking in drinking water overnight, at room temperature	0.023 CB/l and 0.23 CB/l	Freshwater mussel (*Anodontites trapesiali*)	- Burrowing behaviour of tested subjects was atypical, indicative of sub-lethal effects caused by CB leachates.	Montalvão et al. 2019a
Smoked cigarette butts (SCB), smoked cigarette filters (SF), and unsmoked cigarette filters (USF)	Soaking in dilution water for 24 h	SCBs: 0.125 to 4 CBs/l; SF: 0.125 to 8 CBs/l; USF: 0.5 to 16 CBs/l	Topsmelt (*Atherinops affinis*) and fathead minnow (*Pimephales promelas*)	- SCBs showed LC50 (96 h)[b] of 1 CB/l for both species; - SFs showed EC50 (96 h)[c] of 1.8 CB/l (topsmelt) and 4.3 CB/l (fathead minnow); - USFs showed LC50 (96 h) of 5.1 CB/l (topsmelt) and 13.5 CB/l (fathead minnow).	Slaughter et al. 2011
Smoked and unsmoked tobacco (ST and UST); smoked and unsmoked cigarette filter (SF and USF)	Soaking in embryo solution with gentle stirring, for 24 h	0.2 to 20 pieces/l	Medaka embryios (*Oryzias latipes*)	- Low concentrations (0.2 to 5 pieces/l) affected embryonic development and caused anxiety-like behaviour; - High concentrations (10 and 20 pieces/l) caused high lethality;	Lee and Lee 2015

Table 1 (continued)

CB type	Extraction methods	Concentration range	Tested organisms	Main results	Ref.
				- Unsmoked cigarette filter did not cause harm, even with 20 pieces/l concentration.	
Regular cigarette butts, smoked without remnant tobacco; unsmoked	Soaking in solution for 24 h	0.625 to 10 g/l	*Hymenochirus curtipes* *Clarias gariepinus*	- Smoked CBs showed LC50 (96 h) of 1.92 g/l, for both species; - Unsmoked CBs showed LC50 (96 h) of 4.80 g/l, for both species.	Lawal and Ologundudu 2013
Regular cigarette butts, smoked with at least 0.5-cm remnant tobacco	Soaking in tap water, for 1 to 10 days to produce different solutions	0.1 CB/l Days of exposure ranged from 1 to 10	*Aedes albopictus*	- Microcosms with CBs tended to cause more egg deposition as decomposition over time progressed; - Mortality rates of young larvae were significant in microcosms with CBs; - Larvae in late developmental stages were unaffected by CBs.	Dieng et al. 2011
Regular cigarette butts, artificially smoked	Soaking in seawater, for 2 h, stirred every 20 min	0.25 to 5 CBs/l	Snail species *Austrocochlea porcata*, *Nerita atramentosa*, and *Bembicium nanum*	Mortality: after 24 h, 5CBs/l concentration yielded 100% mortality to *Austro* and *Nerita*, while *Bembic* showed 0% mortality. Other concentrations did not cause death in 24 h, for all three species; Behaviours: responses were different among the three species, regarding sub-lethal behavioural changes, with possible implications to snail population dynamics.	Booth et al. 2015
Mentholated cigarette butts, naturally smoked, with a small amount of unsmoked and burnt tobacco	Soaking in deionised, distilled water, for 48 h	0 to 4 CBs/l	Woodland snail *Anguispira alternata*	- No mortality occurred during the 32-day study; food consumption and snail growth did not differ among the effluent concentrations; - During the first week, snails preferred to remain in a 0 CB habitat than in habitats with 1, 2, and 4 CBs; this pattern largely disappeared after 3 weeks.	Gill et al. 2018
Regular cigarette butts, smoked and without remnant tobacco	Soaking in solution, with orbital shaker, for 24 h	0.5 to 8 CBs/l	Ragworm *Hediste diversicolor*	- 8 CBs/l was the lowest concentration that resulted in a significant effect in relative growth rate for CB leachate; - Exposure to 4 and 8 CBs/l leachates prevented burrowing capacity of all individuals during 96 h observation; - Leachate concentration of 8 CBs/l caused significant DNA damage.	Wright et al. 2015
Cellulose fibres from artificially smoked regular cigarettes	Attaching of the fibres to resistors, placed inside nests	Information not available	Parasites of house sparrows (*Passer domesticus*) and house finches (*Carpodacus mexicanus*)	- Parasites tended to move preferably to heat traps containing non-smoked fibres than those containing smoked fibres, presumably due to the presence of nicotine	Suárez-Rodríguez et al. 2013
Cellulose fibres from artificially smoked regular cigarettes	Information not available—nests	Information not available	House finches (*Carpodacus mexicanus*)	- Positive correlation between proportion of CBs in nest and hatching, fledgling, and chick	Suárez-Rodríguez and Garcia 2014

Table 1 (continued)

CB type	Extraction methods	Concentration range	Tested organisms	Main results	Ref.
	evaluated for presence of CBs			immune response of house finches (factors considered beneficial). - However, measurements of genotoxicity in blood cells were also correlated with CB presence in nests, indicating a potentially long-term harm to the species.	
Regular cigarette butts, naturally smoked, with ~ 1-cm remnant tobacco	Soaking in drinking water, for 24 h	Nicotine concentrations: 1.9 μg/l and 1900 μg/l, 60-day exposure	Swiss female mice	- Treatments did not cause locomotor, visual, auditory, and olfactory deficit in mice, and behaviour in test sessions with and without the presence of a snake was not impaired; - All test groups showed defensive behaviour in test sessions with the presence of a cat, but there was a deficit in response for animals exposed to pollutants.	Cardoso et al. 2018
- Regular cigarette butt (RCB); - Menthol cigarette butt (MCB); - Electronic cigarette butt (ECB), after removal from e-cigarette	Soaking in FETAX[d] solution for 1 h	0 to 4 CBs/l (regular and menthol) 0 to 10 CBs/l (electronic)	South African clawed frog *Xenopus laevis*	- RCBs showed LC50 (96 h) of 1.2 CBs/l and EC50 (96 h) of 0.90 CBs/l. - MCBs showed LC50 (96 h) of 1.14 CBs/l and EC50 (96 h) of 0.50 CBs/l. - ECBs showed LC50 (96 h) of 14.6 CBs/l and EC50 (96 h) of 16.3 CBs/l.	Parker and Rayburn 2017

[a] As defined by the European Union Commission decision no. 396/2005, 2009. Please refer to Selmar et al. 2018 for further information

[b] LC50 is the concentration of CB that results in 50% mortality in a given time period (in this case, 96 h). Please refer to Slaughter et al. 2011 for further information

[c] EC50 is the concentration of CB that results in 50% of test organisms show a defined effect (in this case, immobilisation), in a given time period (in this case, 96 h). Please refer to Slaughter et al. 2011 for further information

[d] Frog Embryo Teratogenesis Assay–Xenopus (a toxicity test system that uses embryos of the South African clawed frog). Please refer to Parker and Rayburn (2017) for further information

The potential for heavy metal leaching from discarded CBs was addressed elsewhere (Moriwaki et al. 2009; Moerman and Potts 2011; Dobaradaran et al. 2017; Dobaradaran et al. 2018), and CBs have been demonstrated to be also a source of polycyclic aromatic hydrocarbon (PAH) contamination to the environment (Dobaradaran et al. 2019; Dobaradaran et al. 2020). The changes in metal concentrations due to exposure to CBs also affect the diversity and distribution of microbial populations present in coastal sediments, as reported by Quéméneur et al. (2020). Even if CBs are the focus of this article, it should be noted that recent research by Mansouri et al. (2020) determined that cigarette ash can be more toxic than CBs indicating the importance of controlling this waste as well.

Araújo and Costa (2019) described detrimental aspects of CB littering and accumulation in coastal areas in their recent review article, as well as information regarding the effectiveness of campaigns and actions towards CB pollution awareness and reduction. Another article (Loizidou et al. 2018) attempted to answer the question "Are regular beach clean-ups by local authorities efficient?", and showed that for nine beaches located in Cyprus, even after authorities cleaning procedures, small pieces of litter (including CBs) still remained on the beach.

The productivity of farm crops could be hindered by CB toxicity, as described by Green et al. (2019). These authors reported that littered CBs caused reduction in shoot length and germination success of both perennial ryegrass and white clover, with potential harm to short-term productivity of these species. Food contamination is also possible. In fact, Selmar et al. (2018) reported that inappropriate disposal of CBs in agricultural fields by farm workers cause a substantial contamination of vegetable production due to nicotine uptake by plants. The new pathway of pollution was recorded with a single CB per square metre, with detectable alkaloid leaching into the soil and subsequent uptake by different crops. In this paper, we focus on adverse effects on life forms, including potential indirect damage to human species.

The ubiquity of CB litter is responsible for turning CB into a new waste stream. As informed by Rebischung et al. (2018), the European List of Wastes does not have a specific entry for cigarette butts; instead, it is attached to an entry defined as "Municipal wastes including separately collected fraction/separately collected fraction/other fractions not otherwise specified". The growing conscience about CB toxicity highlights the need for determining which of the fifteen hazardous properties (HP)—a series of hazard classes according to the effect caused, defined in the Waste Framework Directive (EU 2008)—would apply in order to complete its classification as a separate kind of waste. The authors centred their study on the most relevant hazard characteristics for this residue: HP4 (irritant—skin irritation and eye damage), HP5 (specific target organ toxicity/aspiration toxicity), HP6 (acute toxicity), HP7 (carcinogenic), HP8 (corrosive), HP10 (toxic for reproduction), HP11 (mutagenic), HP13 (sensitising), and HP14 (ecotoxic). Tests were conducted according to applicable regulatory documents for three samples, obtained from CB-collecting companies in France. Two samples were dry, and one was wet. Chemical analyses were also performed to evaluate all known pollutants of CB. Results indicated that samples would be classified as hazardous only according to hazard characteristics HP6 (acute toxicity) and HP14 (ecotoxic). Concerning HP6, the authors state that nicotine is the main chemical responsible for toxicity, since its total content in two samples (0.48% and 0.60%) exceeded the 0.25% threshold for a specific category of substances. For HP14, ecotoxicity tests confirmed CB ecotoxicity, but conversely, a calculation method applied did not classify CB as ecotoxic. Nevertheless, as the authors explain, "results of tests are to prevail", and therefore, CBs are also considered toxic according to this specific regulatory requirement.

Chevalier et al. (2018) addressed the dangers of CB litter, but from a nanoscopic scale. Considering a previous study that showed air transfer of soot nanoparticles from partial combustion of cigarettes into human pulmonary system, the authors imagined that such type of nano-litter could be present in CBs, and contaminate leachates as well. One intrinsic characteristic of nanoparticles is their easiness to adsorb chemicals, such as metals and organic contaminants. This happens because of their small size and high surface area. In addition, due to its dimensions, it is a type of contaminant that is not easily contained by physical barriers. Using a mixture of 10 ml of synthetic rainwater and four CBs per vial, samples were put in a rotator-stirring machine under certain conditions for seven full days. After appropriate filtering and concentration of retentates, samples of both smoked and unsmoked CB leachates were analysed via different high-resolution techniques to measure important parameters such as size distribution, aggregation information, and fractal dimension. Total metal and organic carbon percentages were determined for these solutions. After conducting an ultra-filtering process, the authors found that a number of contaminant chemical elements are mostly present in the nano-scale particles than dissolved in the solution; this means that a simple chemical analysis of a filtered solution might not be representative of the real contamination potential of such a solution. Therefore, it is clear that the threats regarding nano-litter particles from CBs are of utmost importance and must be further studied in the future.

Micevska et al. (2006) were amongst the pioneers to evaluate the toxicity of CBs to marine life forms. Two species were subjected to different CB leachate solutions, with varied brands, tar content, etc. Results indicated the acute toxicity that CB impairs to these organisms. In fact, concentrations as low as 25.9 mg/l and 832 mg/l were proven to be toxic to the cladoceran *Ceriodaphnia* cf. *dubia* and bacterium *Vibrio fischeri*.

More recently, Montalvão et al. (2019b) studied how another invertebrate species—freshwater mussel (*Anodontites trapesiali*)—responded to CB litter contamination, finding evidence of mutagenic alteration in the subjects confirmed by results of bioaccumulation of chemical elements. In another article (Montalvão et al. 2019a), the same research group reported behavioural changes in the same species after 14 days of exposure to CB contamination.

Slaughter et al. (2011) were the first to study toxicity aspects of CB leachate samples to vertebrates—in this case, fish, which are ecologically important organisms, often considered indicators of healthy aquatic biomes. A marine species (topsmelt—*Atherinops affinis*) and a freshwater species (fathead minnow—*Pimephales promelas*) were considered test subjects in this study. In addition, different sample conditions were considered—smoked cigarette butts (smoked filter + remnant tobacco), smoked cigarette filters (no tobacco), and unsmoked cigarette filters (no tobacco). Leachates were prepared to concentrations ranging from 0.125 to 16 CBs per litre of solution, which was either dilution water (for fresh water testing) or natural seawater. The LC50—the concentration of CBs that causes 50% mortality—was calculated for each condition considering 96 h. As expected, smoked CBs showed the highest mortality rates, because of toxic chemicals present in remaining tobacco. In this condition, the LC50 was roughly 1 CB/l of solution. By removing this remnant tobacco from the smoked CB, LC50 increased to 1.8 CBs/l (topsmelt) and 4.3 CBs/l (fathead minnow); finally, the unsmoked CB resulted in a LC50 of 5.1 CBs/l (topsmelt) and 13.5 CBs/l for the fathead minnow. Comparing these values with previous mortality rates reported in the literature, the authors show that species such as *Ceriodaphnia* cf. *dubia* and *Daphnia magna* (aquatic invertebrates) are more susceptible to tobacco and CB toxicity than the tested fishes, with toxicity apparent at 0.08 to 0.25 CBs/l.

Lee and Lee (2015) found corresponding results when verifying the developmental toxicity of CB leachates in rice fish embryos (*Oryzias latipes*). While low concentrations of leachates from unsmoked and smoked cigarette tobacco, as well as smoked filters, caused raised heart rate, accelerated development, and changed behaviour, high concentrations caused reduction in heart rate, suppression of development, and increased mortality. Testing with unsmoked cigarette filters showed no effect, even at the highest concentration, also suggesting that CB toxicity derives from tobacco plant itself and from the by-products of its combustion.

Lawal and Ologundudu (2013) conducted another comprehensive study about CB leachate toxicity. They observed the lethality of both smoked and unsmoked CBs to two aquatic species: the western dwarf clawed frog (*Hymenochirus curtipes*) and the North African catfish (*Clarias gariepinus*). Standard 10 g/l solutions were prepared and allowed to rest for 24 h. After filtering, CB leachate solutions had their physicochemical properties evaluated, such as temperature, pH, and heavy metal concentration. For the bioassay, researchers produced five effluent concentrations from the original solutions, ranging from 6.25 to 100% of stock solution concentration. Mortality rates were verified after 24 h, 48 h, 72 h, and 96 h of exposure. Results from physicochemical analyses showed significant contents of heavy metals in both solutions, higher than the Federal Environmental Protection Agency (FEPA) acceptable limits for effluents for all tested elements, except cadmium and copper. Characteristics as odour, colour, dissolved oxygen, and pH range were found to be also inappropriate. The authors reported significant differences between the mortality of unsmoked and smoked CB leachates. Indeed, after 24-h and 48-h exposure, smoked CBs were found to be six times more toxic than their unsmoked counterpart; this difference dropped to 2.5 times more toxicity at the end of the experiment (96-h exposure).

In 2011, Dieng et al. (2011) studied the response of mosquito larvae (*Aedes albopictus*) to the presence of CBs regarding ovipositing (the laying of eggs), larval eclosion, and development. Results indicated that females did not avoid CB-rich habitats to lay their eggs in comparison to the control environment, but CB presence heavily influenced the mortality of young individuals (in early larvae development stages). These findings indicated the potential use of CB as vector control material, since it would attract the deposition of eggs and kill young larvae. Indeed, a 2013 article published by the same group of researchers (Dieng et al. 2013) addressed this question, as will be discussed further in this review paper.

Molluscs that dwell in aquatic environments may also be affected by CB leachate toxicity. Booth et al. (2015) evaluated the impacts of CB leachate on three different species of tide pool snails, i.e. snails that live near the shore and remain for a significant part of their life in tide pools. A CB leachate solution was produced by placing five CBs per litre of seawater, and letting it soak for 2 h. After removing all particles from the solution, it was then diluted at 5%, 10%, 25%, and 100% concentrations. The authors observed behaviour and activity of the snails after different time intervals. Results showed 100% mortality of all species after 8-day exposure at 100% leachate concentration (5 CBs/l), confirming the lethal toxicity of CB leachates. Overall, when compared to other studies involving aquatic life forms, these snails seemed to have more resistance to CB toxicity. Exposure to lower concentrations, or shorter time intervals, exhibited variance in response of each species of snail activity and behaviour. This difference in susceptibility among snail species indicates that high CB pollution may alter community dynamics by differentially affecting certain prey, and thereby predator-prey interactions.

Interestingly, snails with terrestrial habits seem to suffer less from CB toxicity, as described in a paper by Gill et al. (2018). In their first experiment, soil habitats moistened with leachate solutions of equivalent CB concentrations ranging

from 0 to 4 butts/l yielded zero mortality after 32 days, and no difference in behaviour (namely, food consumption and shell growth) was observed. In the second test, snails were placed in a round chamber, with each quadrant containing different quantities of CBs directly placed on the soil. Snails could then choose where to stay, and results showed an initial preference for the control compartment (no CBs), but after day 20 the preference for this area was not significant. These results suggest that leaching of toxicants may occur in a less pronounced manner in land than in aquatic environments.

Research by Wright et al. (2015) about the harmful effects of CB poisoning in ragworms indicated a similar trend. The authors wanted to evaluate how toxic CBs could be to the polychaete worm (*Hediste diversicolor*), an important benthic organism. The potential harm of nicotine to these organisms was tested in two conditions. In the first, CB leachate solutions were prepared and added to the specimens' habitats in different concentrations, promoting a transdermal nicotine uptake, while in the second, solid fragments of smoked CBs were added to soil sediments, and nicotine poisoning occurred through ingestion. Results showed that leachate solutions caused impairment in burrowing times, weight loss, and DNA damage, whereas the experiment using solid fragments did not cause significant effects.

CB littering can affect natural behaviours of species that live or adapted to live in urban areas, such as birds, for example. Birds sometimes incorporate unusual materials in their nests, including man-made materials, often with no apparent reason (Igic et al. 2009). These materials may act as defence mechanisms, to repel potential parasites or to stimulate the offspring's immune system. As highlighted in the work of Suárez-Rodríguez et al. (2013), there is evidence that smoked CBs are becoming one such defence mechanism. The cocktail of chemicals trapped in smoked CBs is actually an advantage in this case, because it can affect distinct parasites that are already resistant to traditional chemical products. Using heat traps to attract common bird ectoparasites, the authors found that the trap containing smoked CBs was less accessed than the trap with unsmoked CBs, presumably due to nicotine presence. In this sense, it is shown that incorporating CBs into nests attends one of the conditions necessary to be classified as self-defence mechanism: it is deleterious to parasites. Later, Suárez-Rodríguez and Garcia (2014) found a positive correlation between the CB proportional usage in nest building and hatching success, fledging success, and chick immune response. The findings indicate a positive effect from CB addition to the nests, but signs of genotoxicity also increased when more butt cellulose was present in the nest. Recently, Suárez-Rodríguez and Garcia (2017) identified that the addition of cigarette butts in these birds' nests occur as a response to the presence of ectoparasites, and not simply because of a resemblance between CBs and feathers.

Mammals are not free from CB contamination. According to a study by Cardoso et al. (2018), water contamination involving cigarette residues can alter the anti-predator behaviour of female Swiss albino mice. This is an important characteristic (anti-predator response) because it relates directly to the central nervous system, since survival when facing a predator is a primitive instinct. Leachate solutions were prepared at two different nicotine concentrations, one considered an environmental level (1.9 μg/l) and one as a concentrated solution (a thousand times more concentrated than the environmental level). Animals were fed these solutions for 60 days, and their behaviour was tested in various ways, starting on day 55. These tests included an open-field test, a locomotion test induced by a light source; olfactory and auditory tests; and, finally, their response to the presence of predators (snakes and cats). Overall, the presence of nicotine did not influence eating or drinking behaviour of test subjects, and no visible signs of intoxication were found. Locomotor activity and auditory and olfactory senses, which are all simultaneously involved in a predator-response situation, were also found to be unaffected in individual tests. However, when considering the presence of a predator, responses varied within sample groups. Animals subjected to nicotine intoxication remained for shorter times in the safe area and had higher frequency upright and higher frequency freezing, evidencing an impaired response to a predator. The authors did not observe a dose-dependent relation, meaning that even at low concentrations, nicotine can affect negatively structures associated with fear responses, at biochemical or molecular level.

A recent class of products which might contribute to reduced CB litter worldwide is electronic cigarettes. These products convert a nicotine solution into a vapour that is inhaled by the customer, and therefore produces no CBs. The absence of this contaminant medium (CBs) is a major advance regarding environment toxicity, but the toxic aspects of the nicotine solution used in these products must be also considered as well as the correct disposal of these electronic devices. Parker and Rayburn (2017) studied the toxicity of disposable electronic cigarettes to aquatic organisms. Embryos of the South African clawed frog (*Xenopus laevis*) were exposed to leachates of regular cigarette butts, flavoured cigarette butts, and electronic cigarette residue. After a 96-h exposure period, with controlled conditions, median values of lethal concentration (LC50), malformation (EC50), non-observed adverse effect concentration (NOAEC), and lowest observed adverse effect concentration (LOAEC) were calculated for each leachate. LC50 and EC50 values obtained for traditional (1.2 and 0.90 CBs/l) and flavoured cigarettes (1.14 and 0.50 CBs/l) were consistent with those reported in previous articles, for other aquatic species, and confirm the acute toxicity of these litter forms to aquatic life. The toxic potential of a disposable e-cigarette, however, was shown to be 10 times lower than traditional cigarettes (even flavoured ones), with LC50 and

EC50 values of 14.6 and 16.31 CBs/l, respectively. Associating this information with the fact that a single e-cigarette is equivalent to approximately 40 traditional cigarettes, the authors state that aquatic environmental impact of e-cigarette butts could potentially be 400 times lower than traditional CBs.

It is worth noting, however, that the aforementioned study focused only on disposable e-cigarettes; it was performed on a sound e-cigarette component without any physical damage that could alter its toxic properties, and is not representative of products that are more complex, with refilling options, for example. In this regard, Baran et al. (2020) considered wastes such as cartridges and e-liquids as well as heat-not-burn tobacco products to evaluate their effects on microorganisms via a commercial multispecies microbial assay for risk assessment (MARA) test. They found that waste from e-cigarettes do not represent a serious threat to the microbiocenosis, while heat-not-burn tobacco product waste showed a strong toxic effect on the investigated microorganisms.

Cigarette butt recycling solutions

Clearly, CB disposal habits and collection/cleaning procedures are crucial to reduce potential toxic contamination of environments, and subsequent harm to a myriad of living species. Nevertheless, if value could be attributed to this litter item, an additional layer of interest would be included which could improve actions towards an ideal scenario of CB disposal and collection. Table 2 reports the different proposed applications for CBs related in this work.

Scientific literature contains a number of propositions to reuse CBs, either in the "as-collected" condition without any pre-treatment or after some kind of treatment. Perhaps, the most direct use of smoked CBs is to produce cellulose acetate, a possibility described by Benavente et al. (2018). Similar research was conducted by Hemamalini et al. (2019), in which cellulose acetate nanofibrous mat was obtained from cigarette filter waste. The following subsections will further discuss other proposed applications of CBs.

Biocide

An interesting application was proposed by Dieng et al. (2013), in an article aimed at verifying the influence of CB chemicals in *Aedes aegypti*, a vector of dengue. Using leachate solutions obtained with deionised water and varying contents of smoked CBs leaching for 24 h, the authors studied the mortality rates of *A. aegypti* in different stages of larvae evolution. A control sample was also produced, having no smoked CBs in the solution. A significant relation between the quantity of smoked CBs in the solution and mortality effect for all larvae development stages was determined. In addition, younger larvae were found to be more sensitive to the presence of toxic elements, for all concentrations tested. Overall, the results showed that CBs have potential to serve as source material for insecticide production aimed at controlling *Aedes aegypti*. Issues that the authors considered material for future studies, at the time, were the potential geno- and cytotoxic effects on humans that these solutions may have as well as their acceptability in regional communities. Interestingly, they also consider that, if decomposing CBs are shown to have the same potential, ashtrays that act as lethal *A. aegypti* traps could be produced.

One year later, the same research group published another study (Dieng et al. 2014) concerning the influence of CB toxicity on diverse behaviours of adult vectors of *Aedes aegypti*. They found that oviposition and larval eclosion rates were not significantly reduced in substrates containing CB leachates. However, results showed reduced fecundity of female adults that developed as larvae in a habitat containing CB leachates, as well as reduced lifespan when compared to control groups. Later on, Murugan et al. (2018) applied this concept to malaria vectors, showing that solutions originated from CBs showed highly toxic properties to young instars and adults of *Anopheles stephensi*, but did not alter the predatory efficiency of *Mesocyclops aspericornis* (a predator of mosquito larvae).

Corrosion inhibition

Zhao et al. (2010a) are among the first researchers to turn to CBs for value, investigating the potential of CB leachate solutions as corrosion inhibitors in N80 steel, a material used in the oil extraction industry. Using a solution consisting of five CBs in 100 ml of distilled water, aged for 24 h, the authors determined the presence of compounds that act in favour of metal corrosion inhibition, by mechanisms described in the original article. Overall, inhibition action relates to the concentration of cigarette butts in water extracts. For a 5 CBs/l solution, maximum inhibition efficiencies of 94.6% and 91.7% in acidified 10% and 15% HCl solutions were found. Additional studies by the same research group found that aerobic CB water extracts are more efficient than anaerobic solutions (Zhao et al. 2010b), and that CuCl addition promotes a synergistic effect to recycled CBs in inhibiting this specific corrosion scenario (Zhang et al. 2011).

Fillers and binders

Lai et al. (2015) studied the incorporation of cigarette butts as a reinforcement in natural rubber, up to 10% in percentage weight, via rolling in a two-roll mill according to ASTM standards. The authors found that the addition of an optimal 4% in weight of cigarette butts to the material improved mechanical characteristics such as tensile strength, tear strength and maximum torque, yielding properties higher than those of the

Table 2 List of studies reporting proposed recycling solutions for cigarette butt management

CB condition	Preparation methods	Proposed use	Main results	Ref.
Smoked butts with 1 cm of unburnt tobacco	Soaking in deionised water	As a novel insecticide product	- *Aedes aegypti* larvae are vulnerable to the presence of CBs in their habitats, and mortality was verified.	Dieng et al. 2013
Smoked butts in diverse conditions	Soaking in aqueous solutions; extraction with ethanol and other solvents	As a source for recovery of cellulose acetate	- Solid cellulose acetate was obtained and characterised.	Benavente et al. 2018
Discarded cigarette butts	Nicotine extraction with methanol, followed by fibre drying	As a source for cellulose acetate nanofibrous mat	- Authors obtained uniform, bead-free cellulose acetate fibre, at 12% weight by volume concentration of cellulose acetate.	Hemamalini et al. 2019
Regular cigarette butts, artificially smoked with remnant tobacco	Soaking in deionised water	As a novel insecticide product.	- CB solutions were attractive to gravid females of dengue vectors, and yielded decreased life span among adults that survived CB exposition.	Dieng et al. 2014
Regular cigarette butts, naturally smoked, with and without remnant tobacco; unsmoked cigarette butts	Soaking in distilled water, boiling, reduction with $AgNO_3$ solution	For the synthesis of Ag nanostructures toxic to malaria vector, malaria parasites and microbial pathogens	- Larvicidal and pupicidal toxicity on malaria vector were verified; - A single treatment with CB extracts and Ag nanostructures reduced significantly the hatchability of malaria vector eggs; - Treatments did not impair the predatory efficiency of a natural predator of malaria vector.	Murugan et al. 2018
Smoked cigarette butts, collected from the road	Soaking in distilled water	As corrosion inhibitor for N80 steel at 90 °C in hydrochloric acid	- Results showed excellent inhibition properties, attaining 94.6% efficiency in 10% HCl solution, with 5% inhibitor.	Zhao et al. 2010a
Used cigarette filters	Extraction of cellulose acetate from CBs, and drying	As reinforcing filler for natural rubber	- The addition of an optimal 4 parts per hundred rubber of cellulose acetate is beneficial in improving rheological and mechanical properties.	Lai et al. 2015
Cigarette butts of different brands and sizes, collected from dry receptacles	Disinfection at 105 °C for 24 h	To be recycled into fired clay bricks	- Inclusion of CBs reduces mechanical properties of fired clay bricks, but there is a firing energy saving in the process; - Proposal: inclusion of 1% CBs in fired clay bricks would not cause significant decrease in	Mohajerani et al. 2016

Table 2 (continued)

CB condition	Preparation methods	Proposed use	Main results	Ref.
			properties, while being effective at getting rid of CBs.	
Cigarette butts of different brands and sizes, collected from dry receptacles.	Disinfection at 105 C for 24 h.	To be recycled into fired clay bricks, in percentages up to 2% in weight.	- Despite mechanical properties reduction, energy during firing process was reduced up to 14%, due to the presence of CBs; - Authors confirmed what was proposed in a previous study—inclusion of 1% CBs in fired clay bricks would not cause significant decrease in properties, while being effective at getting rid of CBs.	Kurmus and Mohajerani 2020
Cigarette butts picked from the roads, without tobacco	Disinfection via sun drying, for 3 days; grinding in mixer	To be incorporated into precast concrete blocks for paving	- Inclusion of 0.1% CBs in weight did not reduce compressive strength of precast concrete blocks significantly, while flexural strength slightly increased. CBs could be included in these percentages (< 0.25% in weight) to obtain good quality precast concrete blocks.	Wadalkar et al. 2018
Cigarette butt waste	Submersion in water and drying for 24 h at 80 °C	As filler for gypsum composites	- Mechanical behaviour improved, except for bond strength which showed no improvement. Acoustic absorption was not altered.	Morales-Segura et al. 2020
Smoked CBs, varied	Drying at 105 °C for 24 h; encapsulation by saturation in hot bitumen	To be incorporated into asphalt concrete for pavement construction	- CBs in 10 and 15 kg/m^3 ratios could be successfully incorporated into asphalt concrete, with acceptable properties.	Mohajerani et al. 2017
Smoked CBs, varied	Oven drying for 24 h; removal of excess tobacco; shredding into fibre using grinder	As fibre modifier in bitumen for the construction of flexible pavements.	- CBs can be used as fibre modifier for bitumen, in order to produce flexible pavements.	Rahman et al. 2020
Used cigarette filters collected from local shop garbage bin	Drying and pyrolysis at 900 °C for 2 h	Production of electrically conducting material	- Obtained pyrolyzed material has good conductive properties, and can be used in different electronic fields while reducing environmental pollution.	Ghosh et al. 2017
Cellulose acetate powder	-	As filler in PVC cement/cellulose acetate composite	- Real permittivity, dielectric loss, and loss tangent increase with increasing cellulose acetate filler content.	Hussein et al. 2017

Table 2 (continued)

CB condition	Preparation methods	Proposed use	Main results	Ref.
Smoked CBs collected from smokers	No chemical modification	To obtain active material for supercapacitor electrode	- An asymmetric supercapacitor produced with this precursor material showed excellent long cycle-life.	Wang et al. 2016
Smoked CBs collected from common areas	Crushing; drying at 80 °C to constant weight	Production of functional carbon precursor for supercapacitors and adsorbents	- Obtained material (with nitrogen doping) showed specific capacitance of 199 F/g (1 A/g) in 6 M KOH electrolyte. Maximum roxarsone adsorption capacity was 679 mg/g, which shows potential for organic pollutant removal.	Li et al. 2020
Smoked CBs	Cutting into small pieces and grinding	To produce nickel oxide@nitrogenous mesoporous carbon nanocomposite for high-performance supercapacitor	- Produced material showed high surface area (~ 1843 m^2/g) as well as specific capacitance of ~ 740 F/g at 1 A/g in 6 M KOH.	Ubaidullah et al. 2020
Smoked CBs	Cleaning and cutting into small pieces	To produce N-doped carbon anode in lithium-ion battery	- Anode made with this material showed satisfactory electrochemical performances.	Yu et al. 2018
Smoked CBs	Separation from coating	As source for promising anode material for lithium-ion batteries	- Anode made with this material presented good cycling stability and rate capability.	Wang et al. 2018
Smoked CBs, collected from public ashtrays	Soaking in distilled water for 24 h; sonication in ethanol for 5 h; drying at 80 °C for 6 h	As source for cellulose acetate, to construct a cellulose-based membrane separator for high-performance lithium-ion battery	- Cellulose-core fibrous membrane displayed properties better than commercial separators and potential to be used in high-power lithium-ion batteries.	Huang et al. 2015
Smoked CBs, collected from public ashtrays	Removal of wrapping paper and cutting into small pieces	To convert into N,S-codoped carbon dots	- As-produced carbon dots were highly fluorescent, with good production yield and a multi-functional nature.	Bandi et al. 2018
Discarded cigarette filters	Washing; drying at 60 °C; mixing with doubly deionised water; autoclaving at 200 °C for 6 h	Production of carbon quantum dots	- As-prepared carbon quantum dots displayed satisfactory results in detecting Sudan I[a].	Anmei et al. 2018
Discarded cigarette filters	Separation from paper and tobacco; soaking and washing with water	For isolating nanocrystalline cellulose	- Cellulose nanocrystals were successfully isolated, with good crystallinity indexes.	Ogundare et al. 2017
Smoked CBs; unsmoked CBs; purchased butts	No pre-treatment	As sound-absorbing material	- High-absorption coefficients were observed from mid frequencies, with performance better than other commercial products measured.	Escobar and Maderuelo-Sanz 2017
Smoked CBs; unused, purchased butts		As sound-absorbing material	- Comparison with commercial fibre materials showed that,	Maderuelo-Sanz et al. 2018

Table 2 (continued)

CB condition	Preparation methods	Proposed use	Main results	Ref.
	Drying at 80 °C for 72 h; separation from paper; shredding into short fibres		for a given thickness, performance is similar or better.	
Used CBs	Separation of filters	To promote catalysis in oxygen reduction reaction	- N-doped carbon support demonstrated to be a promising support material for Pt nanoparticles towards oxygen reduction reaction catalysis.	Kim et al. 2016
Used CBs	No pre-treatment	Preparation of energy storage material for a supercapacitor electrode	- Material exhibited better rate capability and higher specific capacitance when compared to conventional activated carbons.	Lee et al. 2014
Collected CBs	Cut and mixed with deionised water, subjected to 250 °C for 3 h	Preparation of nitrogen-doped CB derived carbon	- Obtained carbon showed excellent specific capacitance, good rate performance, and high cycling stability.	Meng et al. 2019
Waste cigarette filters	Removal of wrapping paper and treating with deionised water	Production of hierarchically porous carbon@graphene composites for lithium-sulphur batteries	- Obtained cathode exhibited excellent electrochemical performance, in terms of cyclability over 1000 cycles and superior rate capability.	Xu et al. 2019
Cigarette filters	Several washing procedures; drying for 72 hours; heat treating at 230 °C, in vacuum oven, for 3 h	To obtain activated carbon via KOH chemical activation	- Surface area and pore diameter resulted in good optimum adsorption capacity for methylene blue.	Hamzah and Umar 2017
Cigarette butts collected from the city	Drying; removal of excess tobacco; grinding	To synthesise hydrochar for adsorption of pollutants	- Results indicated the potential of these hydrochars in applications for removal of organic pollutants from aqueous solutions.	Lima et al. 2018
CBs collected from 10 smokers over a period of 2 months	Oven drying at 80 °C for 12 h, impregnation in NaOH solutions	To obtain activated carbon	- Activated carbons produced from fibres subjected to pre-swelling process showed an increase of up to 47% in specific surface area.	Koochaki et al. 2020
Smoked CBs collected from laboratory colleagues	Removal of wrapping paper and remnant tobacco	As source material for the production of a flow reactor	- A turnover frequency of 370 h^{-1} for reduction of 4-nitro-phenol was reported, with great potential for water treatment.	Xie et al. 2020
Smoked CBs	Not informed	As source of carbon to produce niobium carbide/carbon	- A 54.38% photodegradation of methylene blue dye	Gupta and Pandey 2018

Table 2 (continued)

CB condition	Preparation methods	Proposed use	Main results	Ref.
		nanocomposite, for photocatalyst activity	was reported in 8-h illumination.	
Used and unused CBs	Removal of wrapping paper; grinding	To produce porous carbons with ultra-high surface area	- Hydrogen storage capacity reported was the highest reported to date (at the publication of the paper).	Blankenship and Mokaya 2017
Used CBs	Separation of the filter part; removal of wrapping paper; cutting	To be converted into sensor material for an antihypertensive drug	- Recovery values from 100.25 to 101.25% proved successful application for real sample analysis. Developed sensor could be usable, with excellent stability and repeatability variation parameters.	Bilge et al. 2019
Cigarette filters	No pre-treatment	As a component of a composite aimed at adsorption of phenylbutazone from aqueous media	- Resulting composite showed remarkable properties as an adsorbent for phenylbutazone. A maximum adsorption capacity of 102.22 mg g^{-1} was obtained, suggesting suitability for use in environmental and analytical studies. Satisfactory thermal stability and resistance to leaching was also observed.	Nascimento et al. 2016
Discarded CBs, collected from smoking areas	Soaking in water; cutting of burned tips; repeated washing; drying	For extraction of cellulose nanocrystals and nanofibres, to be used in removing diclofenac from water	- Synthesised material showed diclofenac maximum removal capacity of 107.90 mg g^{-1}. Celluloses extracted from discarded CBs offer promising potential for diclofenac removal from water.	Abu-Danso et al. 2019
Smoked cigarette filters	Carbonisation at 500 °C for 2 h under argon atmosphere; impregnation into KOH solutions	To produce CB based activated carbon, aiming removal of ciprofloxacin (CIP) and sodium dodecyl sulphate (SDS) from aqueous solutions	- Activated carbon reached specific surface area values of 2726 m^2/g. CIP adsorption process was shown to be spontaneous, while SDS showed non-spontaneous adsorption.	Zhang et al. 2020b
Discarded cigarette filters	Removal of remnant tobacco; removal of wrapping paper; grinding	To produce activated carbon for radionuclides adsorption	- Resulting material showed adsorption capacity of 106 mg/g, much higher than that of untreated activated carbon (31 mg/g), with very desirable adsorption	Pu et al. 2019

Table 2 (continued)

CB condition	Preparation methods	Proposed use	Main results	Ref.
			selectivity. Activated carbon material derived from discarded CBs may be a potential adsorbent for removing U(VI)[b] from nuclear wastewater.	
Collected CBs	Washing with ethanol and water several times	To obtain highly porous carbons for capture and reversible storage of CO_2 and volatile iodine	- Resulting porous carbons showed excellent iodine uptake as well as high CO_2 adsorption capacity, both among the highest values reported up to the date when the article was published. Porous carbon from CBs exhibit great potential in reversible adsorption of radioactive iodine and CO_2.	Sun et al. 2017
Collected smoked cigarette butts	Separation from wrapping paper; washing with deionised water; drying	To obtain mesoporous carbonaceous materials for phenol adsorption and CO_2 capture	- Obtained materials showed considerable performance for phenol adsorption and CO_2 capture, indicating a new way to recycle CBs.	Chen et al. 2015
Used cigarette filters	Direct pyrolysis without any pre-treatment	To produce charred carbon for lead removal from aqueous solution	- Removal efficiencies of obtained materials are within acceptable range when compared to other adsorbents aimed at removing lead from aqueous solutions.	Soltani et al. 2015
Smoked and unused cigarette filters	No pre-treatment	To produce selective adsorbent of tobacco-specific nitrosamines	- Obtained sorbents presented high adsorption capability of mesopore and shape selectivity of micropore.	Li et al. 2019
Smoked CBs collected from restaurant smoking room	Washing with ethanol	As a component to produce an electroreduction cell for hexavalent chromium removal from groundwater	- A maximum removal efficiency of 99.7% was observed at a flow rate of 5 ml min^{-1} and current of 0.05 A.	Dorosti et al. 2020
Cigarette filters	Cleaning; shredding; ultrasonication with ethanol	As a raw material for fabricating superhydrophobic and oleophilic material for oil-water separation	- Fabricated filters showed oil sorption capacities considered satisfactory, without significant deterioration after 20 cycles of reuse.	Ifelebuegu et al. 2018
Collected CBs after use	Ultrasonical treatment in ethanol; drying	As a raw material for fabricating superhydrophobic and superoleophilic material for oil-water separation	- Samples yielded 98% efficiency in separation of free oil-water mixtures, using the driving force of gravity.	Liu et al. 2015
Collected cigarette filters	Ultrasonication in ethanol; immersion in NaOH solution; successive ultrasonication in ultrapure water and ethanol	As source material for producing superhydrophobic fibres for oil spill clean-up	- In a staged spill of kerosene in water, prepared material presented 96% clean-up efficiency, without decrease after 10 cycles of use.	Ou et al. 2016

Table 2 (continued)

CB condition	Preparation methods	Proposed use	Main results	Ref.
Collected cigarette filters	No pre-treatment described	To obtain superhydrophobic absorbent for oil spillage clean-up, in association with other chemicals	- As-obtained material retained absorption capacities of 80% (pump oil) and 82% (silicone oil) after 10 cycles.	Xiong et al. 2018
Cigarette filter collected in daily life	Peeling off; washing with hot water several times	To produce nanofibrous membranes for immiscible oil/water mixtures and emulsions separation	- Obtained material (cigarette filter coated meshes) showed separation efficiency larger than 99.9% for immiscible oil/water mixtures and emulsions after many cyclic testing.	Liu et al. 2019
Collected CBs after use	Ultrasonical treatment in ethanol; drying	As source material for oil-water separation, after coating with polypyrrole	- Material showed excellent oil/water separation performance. Efficiencies reached 97.3% and 99% for light and heavy oil, respectively; good stability was also obtained. Absorption capacities reached 9.4–22.7 times the mass of the filter.	Zhang et al. 2020a
Collected waste cigarette filters	Removal of wrapping paper; rinsing	To obtain graphene aerogel composites for electromagnetic wave absorption	- Obtained composite aerogels showed strong electromagnetic wave absorption abilities and wide absorption bandwidths.	Wang et al. 2015
Smoked cigarette filters	Removal of wrapping paper	To produce ester-rich bio-oil through microwave-assisted pyrolysis	- A 29.17% bio-oil yield was obtained from microwave-assisted pyrolysis of CBs; this yield was raised to 46.71% by adding methanol into nitrogen carrier gas during the process.	Wang et al. 2020
Collected cigarette filters	Washing with ethanol and distilled water, several times	To produce photothermal conversion material for solar steam generation	- Authors reported energy conversion efficiency of 94% under stimulated solar illumination.	Sun et al. 2019
Cigarette butts from collectors in restaurants, bars and buildings	No pre-treatment	As source material for recycling into cellulose pulp	- It was possible to produce cellulose pulp from CB waste, in a process which does not generate liquid effluent, and was licenced to a recycling company.	Teixeira et al. 2017

[a] Sudan I is a synthetic azo-colourant dye, formerly used but now strictly forbidden in several countries. Please refer to Anmei et al. 2018 for further information

[b] U(VI) stands for hexavalent uranium. Please refer to Pu et al. 2019 for further information

control sample. Care must be taken, however, with potential toxicity issues, as considered by the authors.

Mohajerani et al. (2016) envisioned a possible solution involving the civil construction, a branch of industry that relates to voluminous amounts of supplies. To study the feasibility of incorporating CBs into fired clay bricks, the authors evaluated the influence of distinct variables on mechanical properties of the bricks—CB content, mixing time, and heating time. They found that the addition of CBs in the quantities proposed (2.5%, 5%, 7.5%, and 10% in weight percentage) resulted in reduced values of dry density and compressive strength, such that bricks containing more than 2.5% CBs would not be feasible for construction. However, samples containing 2.5% CBs presented only a slight reduction of mechanical properties, attaining requirements for constructing buildings. Ultimately, the authors theorise that the addition of 1% CBs in weight would represent a practical solution to the CB litter problem, since the reduction in mechanical properties would be minimal. Trough simple calculations, they reach the conclusion that the brick production worldwide is so massive that if only 2.5% of the brick production could incorporate 1% in weight of CBs, all of the littered CBs globally would be used. Indeed, in 2020, a new publication by the same group (Kurmus and Mohajerani 2020) investigating CB additions in percent weight from 0.5 to 2% confirmed this hypothesis.

Wadalkar et al. (2018) published work also considering CBs as fillers for the construction industry, but in precast concrete blocks. They found that the addition of 0.1% CBs in weight does not cause significant reduction in mechanical properties of these blocks, and therefore, this application could represent a practical solution to use CB litter. Another construction material that can be used with CB waste is gypsum, as reported by Morales-Segura et al. (2020). Results indicated suitability of use in applications where bonding to a ceramic support is not needed, requiring high surface hardness and/or mechanical resistance.

Asphalt concrete (AC) for pavement construction could also incorporate CBs, as reported by Mohajerani et al. (2017). This use would also benefit from the large-scale amounts involved with automotive transportation. In order to avoid toxicity issues associated with chemical translocation of toxic materials present in the cigarette butts, the authors performed an encapsulation step, prior to AC mixing, consisting of soaking the material in an encapsulating medium. The authors present the results of two studies: firstly, they evaluated the effects of incorporating CBs encapsulated with three distinct classes of bitumen (C170, C320, and C660), in different proportions (10 kg/m^3, 15 kg/m^3, and 25 kg/m^3), into AC mixes manufactured with one specific class of bitumen (C170). Afterwards, the authors assessed the influence of adding 10 kg/m^3 of CBs encapsulated with paraffin wax into AC mixes produced with two distinct classes of bitumen (C170 and C320). In the first study, findings showed that AC fabricated with 10 kg/m^3 or 15 kg/m^3 CBs yielded acceptable properties, such as flow, stability, resilient modulus, bulk density, and percentage of air voids. CBs encapsulated with higher grade bitumen generated ACs with greater bulk density and maximum density, as expected, and the addition of CBs in larger quantities decreases the general density of AC, since CBs replace heavier aggregates in the mix. The second study proved that the addition of 10 kg/m^3 of paraffin encapsulated CBs in both C170 and C320 AC mixes is viable and yields properties acceptable for light traffic requirements, which would be a possible destination for CBs. More recently, Rahman et al. (2020) evaluated the application of CBs in bitumen as a replacement for virgin cellulose fibre as fibre modifier, with very interesting results.

Electric material and batteries

In the field of electric applications, CBs demonstrate interesting possibilities. For example, good conductive material can be obtained by a simple heat treatment process (Ghosh et al. 2017). Hussein et al. (2017) considered CBs potential filler material for a PVC cement resin composite, which is a substance with adhesive properties commonly used to "weld" pieces of plastic together. Commonly used as insulators in voltage engineering, these composites may have their dielectric permittivity and electrical conductivity tuned through incorporation of conductive fillers, such as CBs. For different filler concentrations (0.1%, 0.2%, 0.3%, 0.4%, and 0.5%), the authors studied the material response in terms of dielectric properties when altering frequency range and temperature range. They found that the permittivity increased as filler content and/or temperature increased, due to effects of interfacial polarisation (filler content) and segmental mobility of polymer molecules (temperature); higher frequencies, however, yielded lower values of permittivity. Also, the composite showed lower values of electrical impedance with higher values of filler content, temperature, or frequency.

Examples of research involving batteries are those of Yu et al. (2018) and Wang et al. (2018), who described recovery of CBs for producing anode materials in lithium-ion batteries, and Xu et al. (2019) who produced hierarchically porous carbon@graphene composites derived from carbonised CBs applicable as cathode hosts for lithium-sulphur batteries. Earlier on, Huang et al. (2015) had proposed to use recycled cigarette filters in preparing membranes to act as separators in lithium-ion batteries.

Supercapacitors

Wang et al. (2016) proposed the application of used cigarette filters as parent material for obtaining carbon fibre/vanadium nitride precursors for supercapacitors. Initially, CBs were

subjected to a solution in order to absorb a specific compound (NH_4VO_3), and afterwards, a heat treatment in nitrogen-rich atmosphere was responsible for transforming this compound into vanadium nitride, as well as turning the cellulose fibres into carbon fibres. The obtained carbon fibre/vanadium nitride material was then used to fabricate a supercapacitor device, which showed excellent long cycle-life attaining an 82% capacitance retention rate after 1000 cycles.

Li et al. (2020) reported the production of a functional carbon precursor by direct pyrolysis of CBs for supercapacitors and adsorbents, in which the properties of the material were shown to be strongly affected by the activation process parameters. Using a different processing method, called hydrothermal synthesis, Ubaidullah et al. (2020) obtained a novel nickel oxide@nitrogenous mesoporous carbon nanocomposite, using smoked CBs, with brilliant electrochemical performance in supercapacitors as well as excellent stability up to 4000 segments.

Lee et al. (2014) used smoked cigarette filters to obtain nitrogen-doped meso/microporous hybrid carbon material. Results showed better rate capability and higher specific capacitance for the supercapacitor when compared to conventional activated carbons; a similar work with promising results was published more recently by Meng et al. (2019).

Nanomaterials

Ogundare et al. (2017) studied the feasibility of using CBs as source material for nanocrystalline cellulose, a promising material that presents interesting properties such as high strength, high surface area, low density, low toxicity, and biocompatibility. For this reason, sources for its production have recently included recycled materials. In their work, from discarded CBs, the authors managed to isolate nanocrystalline cellulose. The results of their characterisation were promising, and transmission electron microscopy (TEM) revealed needle-like cellulose nanocrystals measuring 143 nm in length and 8 nm in width, in average, which represents an interesting aspect ratio for use as reinforcement. The overall nanocellulose yield from CBs was 29.4%, which is more than two times the value obtained from old corrugated paper (13.3%) and roughly 1.5 times the value obtained for wastepaper (19.0%). It was concluded that this material is feasible for nanocrystalline cellulose production, if there is an appropriate collection logistics by consumers.

Carbon dots, a new emerging category of nanomaterials, have been evaluated as potential outputs of CB litter research. Bandi et al. (2018) reported the conversion of toxic CBs to carbon dots, which showed a number of potential applications, including security ink, bioimaging, and sensing, while Anmei et al. (2018) envisioned the application of carbon quantum dots obtained from cigarette filters in fluorescence detection of an important carcinogenic food colourant.

Sound absorption

Acoustical isolation could take advantage of fibrous and porous structures of cellulose acetate, the main component of cigarette filters. Porosity plays a very important role in acoustic properties, and these samples could profit from a "double porosity": the inner porosity of the CBs, as well as the porosity resulting from air gaps between packed CBs (Escobar and Maderuelo-Sanz 2017). Following ISO 10534-2:1998 standard, the authors evaluated the acoustical performance of samples prepared by simply placing the CBs inside impedance tubes appropriate for this kind of measurement, without using binders. Different types of CBs were tested (smoked CBs, unused CBs manually separated from the cigarette, and purchased CBs), as well as different packing conditions (10 to 14 butts packed inside the tube). When compared to commercial sound-absorbing materials, samples prepared with CBs showed superior acoustic absorption at mid frequencies (from 500 to 4000 Hz). Thus, the fabrication of novel sound-absorbing materials from CBs is promising.

In the following year, Maderuelo-Sanz et al. (2018) conducted further studies regarding sound absorption. This time, researchers conducted a pre-treatment of CBs consisting of 72 h drying at 80 °C, separation from involving paper, fractioning in short fibres, and mixing, in order to obtain a homogeneous mixture and, thus, measure more reliable physical properties. The mixture was then compacted, without using any binder. In this work, they concluded that sound absorption is higher when bulk density and/or porosity increases, and also when flow resistivity is reduced. The findings corroborate those of the first study, indicating an interesting potential for this material as sound barriers.

Energy and related topics

Kim et al. (2016) envisioned the potential of CBs to function as a supporting material for oxygen reduction reaction (ORR). In fuel cells, electrocatalysts have the function of accelerating the ORR, and the traditionally used material is platinum due to its high activity. However, factors such as high cost and limited storage encourage research towards reducing platinum use and enhancing catalytic activity of platinum. Supporting materials with high surface area, adequate pore structure, and good conductivity, such as carbon materials, are used to promote this catalytic enhancement by permitting high dispersion of platinum nanoparticles. In their work, carbon materials were produced from used cigarette filters, after separation and heat treatments at 900 °C for 2 h, in two conditions: N atmosphere (N-doped carbon) and Ar atmosphere (without doping). Then, platinum dispersion was conducted by the conventional impregnation method. The advantages of this simple route are the recycling of CBs as carbon source for platinum supports, the development of a bimodal structure

(pores in different scales—mesopores and micropores), and simultaneous introduction of N functional groups (in the case of N-doped carbon), which are important to the formation of homogeneous nucleation sites for platinum nanoparticles. Following an extensive set of characterisation methods, it was found that high surface area, bimodal pore size distribution, and N doping contributed to excellent results in terms of platinum dispersion—in fact, dispersion was higher than that of commercially used carbon support material.

Blankenship and Mokaya (2017) published a study proposing CBs as source for ultra-high surface area carbons destined for hydrogen intake. Hydrogen is being researched as an energy source material because of two main characteristics: its gravimetric energy capacity, and the fact that this material does not generate CO_2 emissions when used for energy production. Naturally, there is also a driving need for materials capable of storing this hydrogen, and materials with high surface area (a common trait of porous materials) can store considerable amounts via physisorption. The authors found an interesting relation concerning smoked CBs and the effect of activation temperature on their porosity. Usually, higher activation temperatures yield higher porosities and surface areas, and this trend was observed in unsmoked CBs. For smoked CBs, however, optimal porosity and surface area properties were encountered at an intermediate activation temperature, and higher temperatures yielded lower results. The explanation for this anomaly, according to the authors, is the presence of metal additives (such as K, Ca, Na, and Mg) as remains from the tobacco combustion process. Working as activation sites, they sum up to the activation effect caused by KOH and cause this odd behaviour. Exceptional results of hydrogen storage were reported in this article—in fact, the results are presented as "highest hydrogen uptake capacity reported to date for any carbons or porous materials in general", combining the ecological effects of clean fuel emissions, waste disposal, and valorisation of resources in a perfect manner.

In 2020, Wang et al. addressed the production of ester-rich bio-oil from used cigarette filters via microwave-assisted pyrolysis process, and they estimated that over 350,000 MT of this bio-oil can be produced if the technology used in the process could be scaled up worldwide (Wang et al. 2020).

An application which could contribute twofold to a cleaner environment is the utilisation of CBs to obtain phototermal conversion material, as described by Sun et al. (2019). In this scenario, material produced from CBs and coated with reduced graphene oxide is able to absorb sunlight and convert it into heat, which in turn will generate clean steam energy; indeed, the authors reported energy conversion efficiency of 94% in conditions similar to those practical in solar energy generation.

Chemical adsorption

Hamzah and Umar (2017) proposed the fabrication of active carbons from cigarette filter waste, using KOH chemical activation induced by microwave power input. Surface area and methylene blue adsorption capacity were evaluated with promising results. Lima et al. (2018) also studied the potential of CBs for the adsorption of pollutants, describing a spontaneous methylene blue adsorption process happening in hydrochars fabricated from CBs. The potential for activated carbon production may even be enhanced by pre-swelling of CBs, raising specific surface area up to 47%, as explained by Koochaki et al. (2020). Different approaches related to degradation of pollutants in water include a study where CBs were used as source for a superfast flow reactor (Xie et al. 2020), and an article describing the fabrication of a niobium carbide/carbon nanocomposite to be used as photocatalyst in which CBs were the carbon source (Gupta and Pandey 2018).

Efforts for CB valorisation have been made also in the drug and pharmaceutical industry. For example, Bilge et al. (2019) recently published an article in which toxic CBs demonstrated potential to become sensor material for sensitive determination of an antihypertensive drug. In 2016, Nascimento et al. (2016) investigated the potential application of cigarette filters in a composite designed to work as an adsorbent of phenylbutazone (PBZ), a commonly used anti-inflammatory drug. This pharmaceutical ends up in the environment, and long-term exposure to it may result in adverse effects to humans, as well as wildlife. Besides CBs, this composite also contains polypyrrole (PPy) and polyaniline (Pani), two polymers that as a composite have properties such as hydrophobicity, acid-base character, and polar functional groups. The composite was produced in a single-step triple-phase interfacial reaction, in a temperature from 0 to 5 °C. This simple fabrication process contributes to the feasibility of using the material as an adsorbent. Additionally, properties such as high surface area, heterogeneous surface, and modest porosity are of interest and, indeed, resulted in a maximum adsorption capacity of 100.22 mg g^{-1}—a result that indicates promising future applications in the separation field.

Other examples of drug removal applications include that by Abu-Danso et al. (2019) of using processed cellulose from CBs to absorb diclofenac from water and Zhang et al. (2020b), who produced an interesting CB-based activated carbon material of high surface area to remove ciprofloxacin and sodium dodecyl sulphate from aqueous solutions.

The potential as an absorbent medium was also investigated for the disposal of wastewater generated in nuclear plants, which contain radionuclide uranium (Pu et al. 2019). The adsorption capacity surpassed that of untreated activated carbon and showed a very interesting adsorption selectivity. Earlier, Sun et al. (2017) presented interesting results concerning the adsorption capacity of CBs, after KOH activation, for CO_2 and iodine vapour

capture. Other examples of adsorption studies that use CB waste are the preparation of materials for phenol adsorption (Chen et al. 2015), removal of lead from aqueous solution using charred carbon derived from used cigarette filters (Soltani et al. 2015), and selective adsorption of tobacco-specific nitrosamines in solution (Li et al. 2019). In an application concerning electroreduction, CBs were able to contribute in reducing hexavalent chromium contamination of groundwater by acting as a palladium nanocatalyst immobiliser (Dorosti et al. 2020).

In 2018, Ifelebuegu et al. (2018) reported that surface-modified cigarette filters can act as a superhydrophobic and oleophilic sorbent, suitable for oil spill clean-up applications. This was not the first work to address this possibility. In fact, CBs and derivatives were considered for applications involving a mixture of different oils (Liu et al. 2015), kerosene (Ou et al. 2016), silicone oil (Xiong et al. 2018), and isolated different oils (Liu et al. 2019). Enhancements such as coating CBs with polypyrrole particles modified with dodecanethiol were also reported for oil separation (Zhang et al. 2020a, b), indicating a promising application for this waste.

Electromagnetic wave absorption

Interestingly, electromagnetic wave absorption is another field where recycled cigarette filters may be useful, as published by Wang et al. (2015). Electromagnetic waves are a considerable source of interference and radiation, mostly from wireless technology, digital devices, and radar systems. The authors obtained graphene aerogel composites from CB litter, which showed good results—further enhanced by a polypyrrole coating.

Pulp production

One great example of a complete process for CB recycling is the production of cellulose pulp for paper industry, using CBs and alkaline pulping (Teixeira et al. 2017). Samples prepared with CBs showed satisfactory mechanical properties for use, and a low degree of whiteness that may require additional bleaching depending on the final use of the paper. The by-product of this process is a dark liquor generated from cooking the mixture, which showed eventual presence of aldehydes and other carbonylated compounds, nitrosamines, nicotine, and heavy metals. Under Brazilian law, this by-product could be discarded directly in a water body without further treatment, even if compounds such as nitrosamines and nicotine were detected. This fact occurs because current Brazilian laws do not establish limits for these specific chemicals. Nevertheless, the authors studied possible means of removing this toxicity. They devised a way of incorporating this clarified effluent repeatedly into new cooking batches, as part of the mixture. As explained in the article, the whole process was licensed to a recycling company, which is producing good-quality pulp for a cardboard factory. Since the effluent is reused in the process, this is an example of a complete cycle that effectively contributes to clean our environment of CBs.

Conclusion

Environmental hazard results from reckless improper disposal of toxic waste, and cigarette butts are proven in literature to be source of damage to a variety of living species, from invertebrate diminutive aquatic species to adult rodents, even if present in considerably small concentrations. The knowledge about its contamination power is paramount to enlighten smokers and enhance public efforts aimed at raising worldwide awareness regarding this topic. While the toxic effects of CBs to simpler organisms are more present in literature, there seems to be a gap in relation to research concerning long-term effects of CB toxicity to mammals, indicating a possible future research trend.

Considering the variety of possibilities for recycling cigarette butts described in this article, ranging from applications with zero to low value addition (aimed mostly at getting rid of CBs) to high-value end products derived from litter, it is possible to foresee a future where careless dumping of cigarette filters in the environment will be minimised. Applications related to chemical adsorption, energy production, and creation of nanomaterials seem to be excellent options that add value to this litter and therefore must be further investigated, while the incorporation of CBs in civil construction is interesting considering the significant amount of CBs that could be used. In addition, the search for new applications must continue since the creation of value is the key factor to transform cigarette butts from an unwanted persistent litter to a desired valuable raw material.

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Compliance with ethical standards

Competing interests The authors declare that they have no conflict of interest.

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